

1. What is Django Rest Framework (DRF)?

Django Rest Framework is a **powerful toolkit** used to build **RESTful APIs** using Django.

✦ Key Points:

- Built on top of Django
- Used to create Web APIs
- Supports JSON, XML responses
- Widely used for backend services

🎯 Interview Line:

“DRF is used to build REST APIs using Django.”

2. Why do we use DRF instead of Django?

Django is mainly used for **server-side rendered applications**, while DRF is used for **API-based applications**.

✦ Reasons:

- API support (JSON response)
 - Used with React, Angular, Mobile apps
 - Built-in authentication
 - Better request/response handling
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3. What is REST?

REST stands for **Representational State Transfer**.

✦ REST Principles:

- Stateless communication
- Uses HTTP methods
- Client-server architecture
- Resource-based URLs

✦ HTTP Methods:

- GET → Read
- POST → Create
- PUT → Update
- DELETE → Delete

4. What is an API?

An API (Application Programming Interface) allows **two applications to communicate** with each other.

□ Example:

- Frontend → API → Backend
 - Mobile App → API → Database
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5. What are Serializers in DRF?

Serializers convert **complex Django objects into JSON** and vice-versa.

✦ Key Points:

- Similar to Django Forms
- Validate data
- Convert model instances to JSON

□ Example:

```
from rest_framework import serializers
```

```
class StudentSerializer(serializers.ModelSerializer):
```

```
    class Meta:
```

```
        model = Student
```

```
        fields = '__all__'
```

☞ Interview Line:

“Serializers convert model objects into JSON format.”

6. Difference between Serializer and ModelSerializer

Feature	Serializer	ModelSerializer
Manual fields	Yes	No
Auto fields	✗	✓
Validation	Manual	Automatic
Use case	Custom data	Model-based data

7. What are Views in DRF?

Views handle API logic and return **Response objects instead of HttpResponse**.

✦ Types of Views:

1. Function-Based Views
2. Class-Based Views
3. APIView
4. Generic Views

8. What is APIView?

APIView is a class-based view provided by DRF to handle API requests.

✦ Key Points:

- Supports HTTP methods (GET, POST, PUT, DELETE)
- More control than generic views
- Requires more code

9. What are Generic Views in DRF?

Generic views provide **pre-built CRUD operations**. "Generic views reduce boilerplate code."

- ListAPIView
- CreateAPIView
- RetrieveAPIView
- UpdateAPIView
- DestroyAPIView

10. What is Authentication in DRF?

Authentication verifies **who the user is**.

✦ Types:

- Session Authentication
 - Token Authentication
 - JWT Authentication
 - Basic Authentication
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11. What is Pagination in DRF?

Pagination limits the number of records returned in an API response.

✦ Benefits:

- Improves performance
 - Reduces load time
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12. What is Status Code in DRF?

✦ Common Status Codes:

- 200 → OK
 - 201 → Created
 - 400 → Bad Request
 - 401 → Unauthorized
 - 403 → Forbidden
 - 404 → Not Found
-

13. What is Response object in DRF?

DRF uses Response instead of HttpResponseRedirect.

```
from rest_framework.response import Response  
  
return Response({"msg": "Success"})
```

14. Real-Time Example (Interview Favorite)

□ Online Exam System API:

- GET → View questions
 - POST → Submit answers
 - PUT → Update profile
 - DELETE → Remove question
-

15. What is Postman?

Postman is a **popular API testing tool** used to develop, test, and debug APIs.

✦ Key Points:

- Used to test REST APIs
- Supports GET, POST, PUT, DELETE
- Sends HTTP requests to servers
- Shows response in JSON/XML

🔗 Interview Line:

“Postman is used to test APIs without frontend.”

16. Why do we use Postman?

Postman helps developers **test APIs quickly** without writing frontend code.

✦ Uses:

- Test backend APIs
 - Check request & response
 - Debug API issues
 - Validate authentication
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17. Which HTTP methods are supported by Postman?

✚ Answer:

- GET – Fetch data
 - POST – Insert data
 - PUT – Update data
 - PATCH – Partial update
 - DELETE – Remove data
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18. What is an API request in Postman?

An API request contains all the information needed to communicate with the server.

✚ Components:

- URL
 - HTTP Method
 - Headers
 - Body
 - Params
-

19. What are Headers in Postman?

Headers contain **metadata** about the request.

✚ Examples:

- Content-Type
- Authorization
- Accept

Content-Type: application/json

21. What is Body in Postman?

Body is used to send **data to the server**, mainly in POST & PUT requests.

✚ Body Types:

- raw (JSON)
- form-data

- x-www-form-urlencoded
- binary

22. Difference between Params and Body

Params	Body
Sent in URL	Sent in request body
Used in GET	Used in POST/PUT
Visible	Not visible

23. What is Collection in Postman?

A collection is a **group of API requests** saved together.

🔑 Benefits:

- Easy organization
- Reusable APIs
- Used for automation

24. What is Environment in Postman?

Environment stores **variables** used in requests.

🔑 Example:

```
{{base_url}}
```

🔑 Use Case:

- Dev, Test, Production URLs

25. What is Status Code in Postman?

🔑 Common Status Codes:

- 200 – OK
- 201 – Created
- 400 – Bad Request
- 401 – Unauthorized
- 403 – Forbidden

- 404 – Not Found
 - 500 – Server Error
-

26. What is Response in Postman?

Response is the **output returned by the server**.

🔑 Includes:

- Status Code
 - Response Body
 - Headers
 - Response Time
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27. What is Newman?

Newman is a **command-line tool** to run Postman collections.

🔑 Used For:

- Automation testing
 - CI/CD pipelines
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28. Difference between Postman and Swagger

Postman

API Testing Tool

Manual & Automated

Used by testers

Swagger

API Documentation Tool

Auto-generated docs

Used by developers

29. How do you send JSON data in Postman?

1. Select POST method
2. Body → raw → JSON
3. Add Header:

Content-Type: application/json

30. Difference Between PUT and PATCH

Feature	PUT	PATCH
Type of update	Full update	Partial update
Data required	Entire object	Only changed fields
Risk of data loss	High	Low
Performance	Slower	Faster
Real use	Replace record	Edit record

